

# Korean Petrochemical Industry

## Decarbonization Pathway Analysis

### Marginal Abatement Cost Curve (MACC) Model

Report Generated: 2025-10-12 00:32

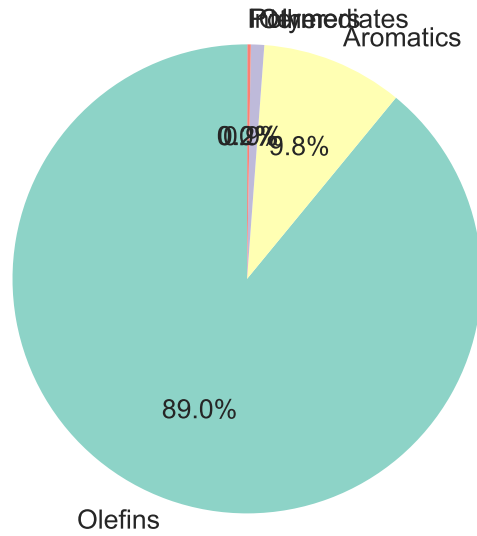
Model Version: 2.2

#### Key Model Parameters

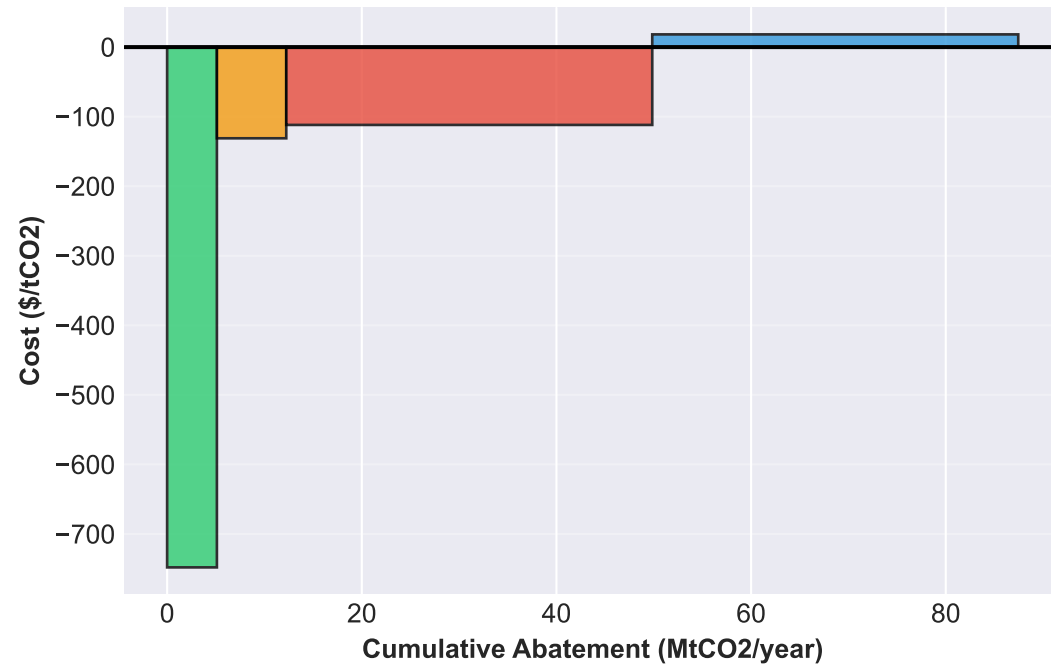
- Baseline Emissions (2025): 52.0 MtCO<sub>2</sub>/year
  - Facilities Analyzed: 248
- Technologies: 4 (Heat Pump, RE PPA, NCC-H<sub>2</sub>, NCC-Electricity)
  - Time Horizon: 2025-2050
- Maximum Abatement Potential: 49.8 MtCO<sub>2</sub> (95.9%)
  - Scenarios: Conservative, Moderate, Aggressive

# Executive Summary

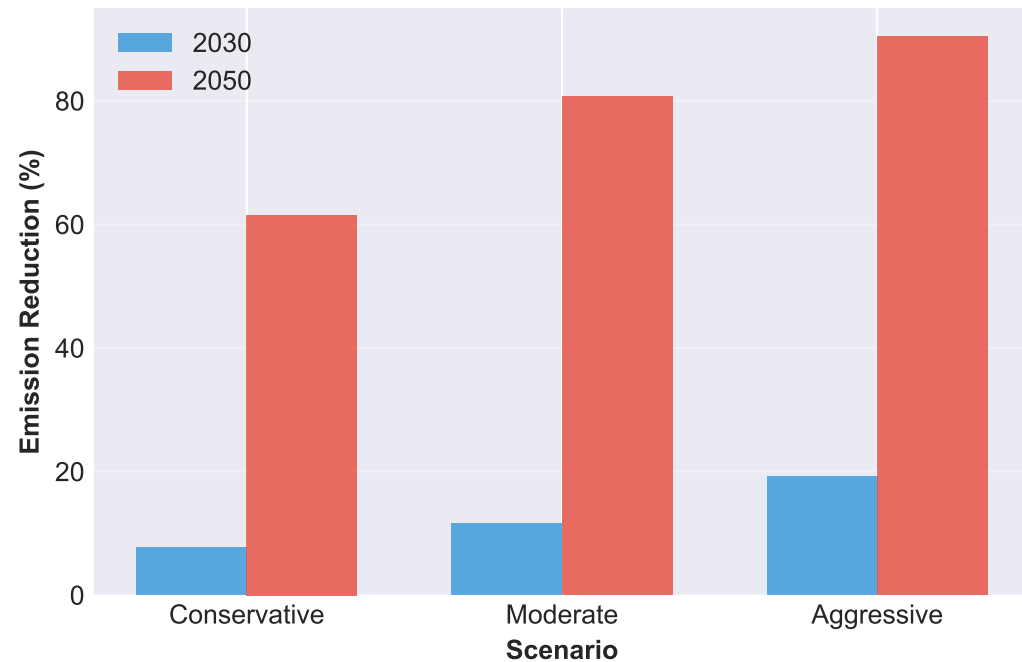
## 2025 Baseline by Product Group (52.0 MtCO2 total)



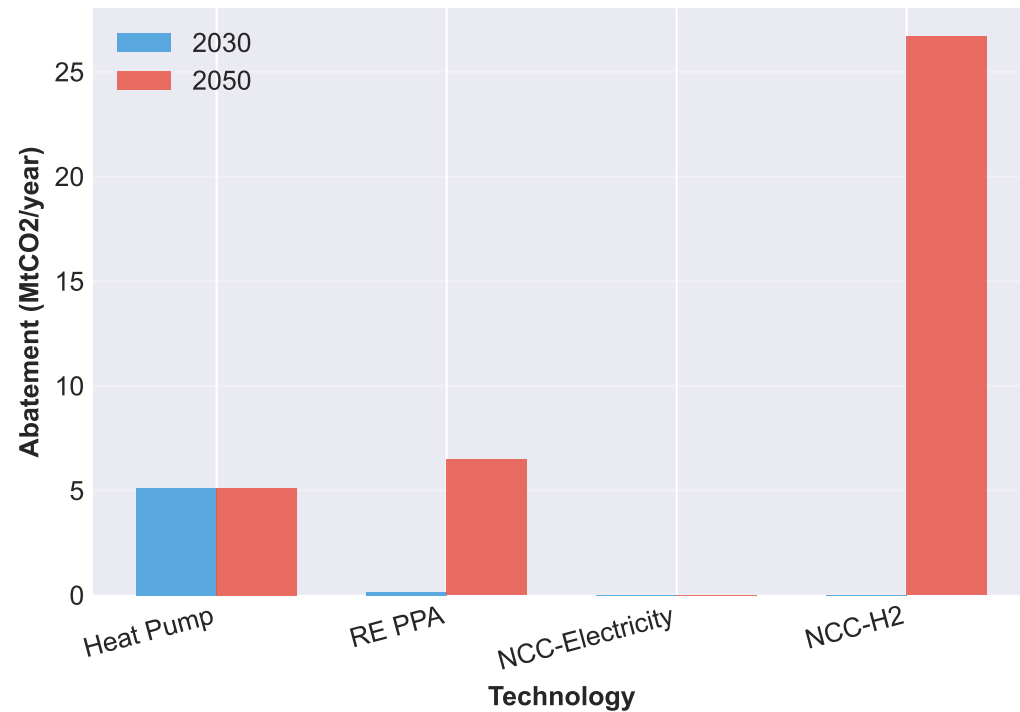
## MACC Curve - 2030 (3 cost-negative technologies)



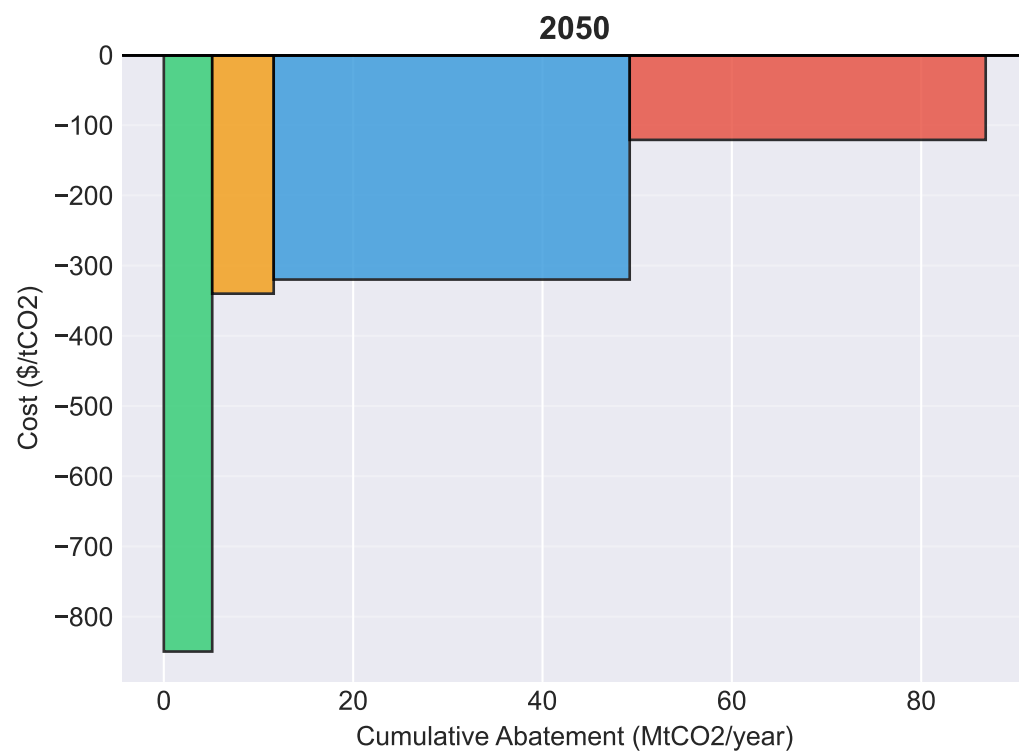
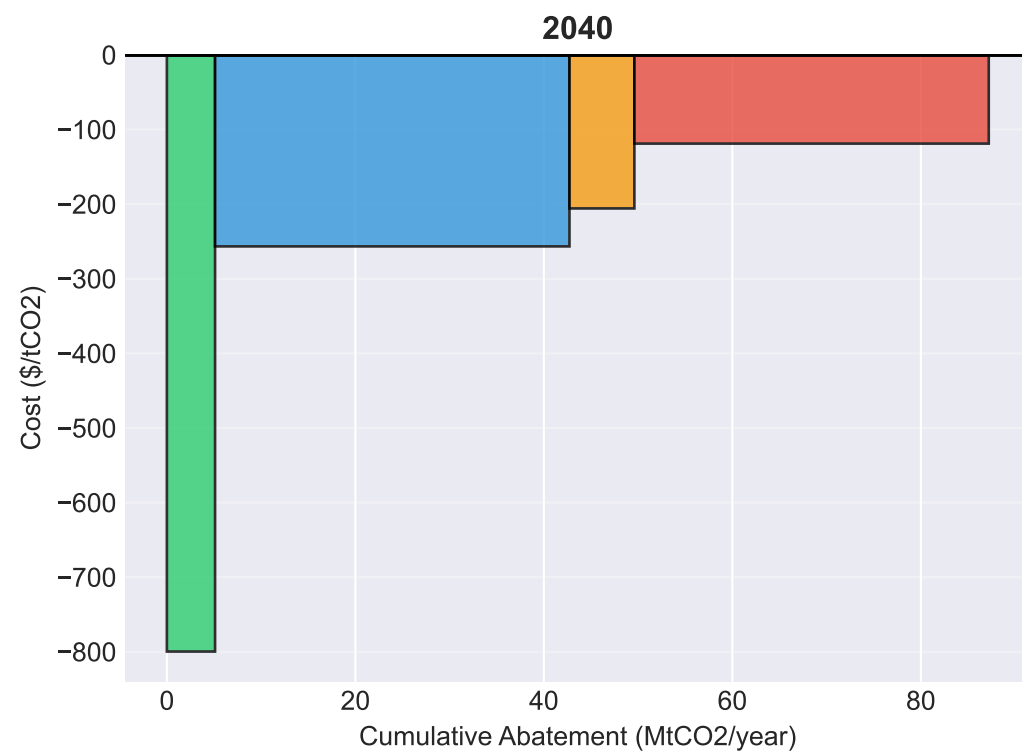
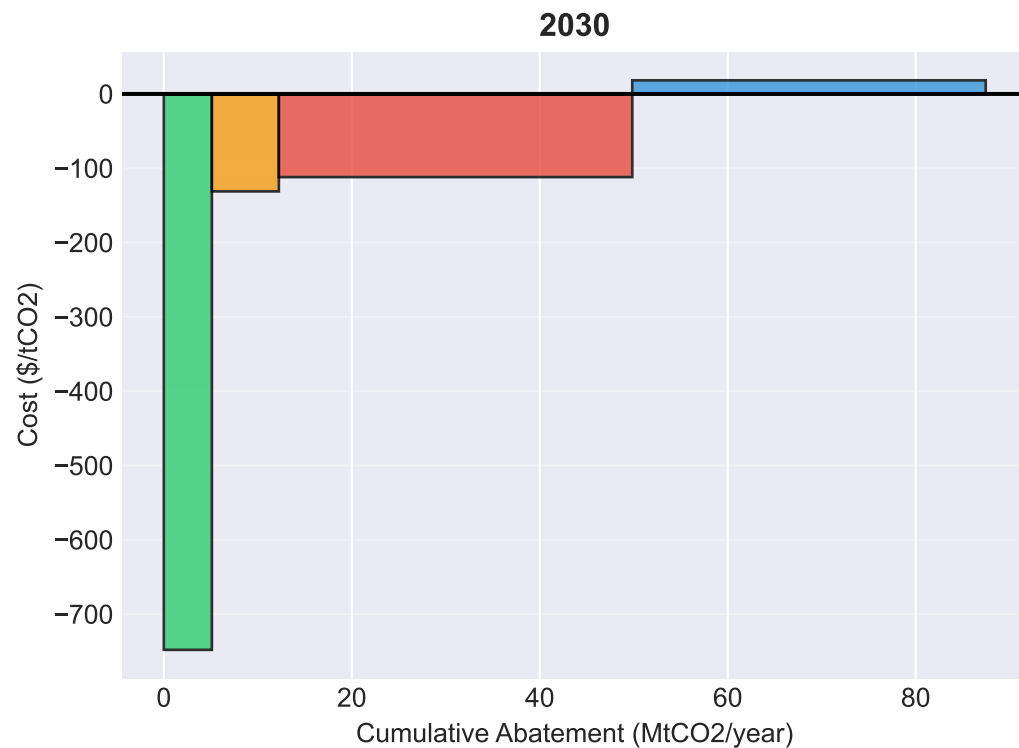
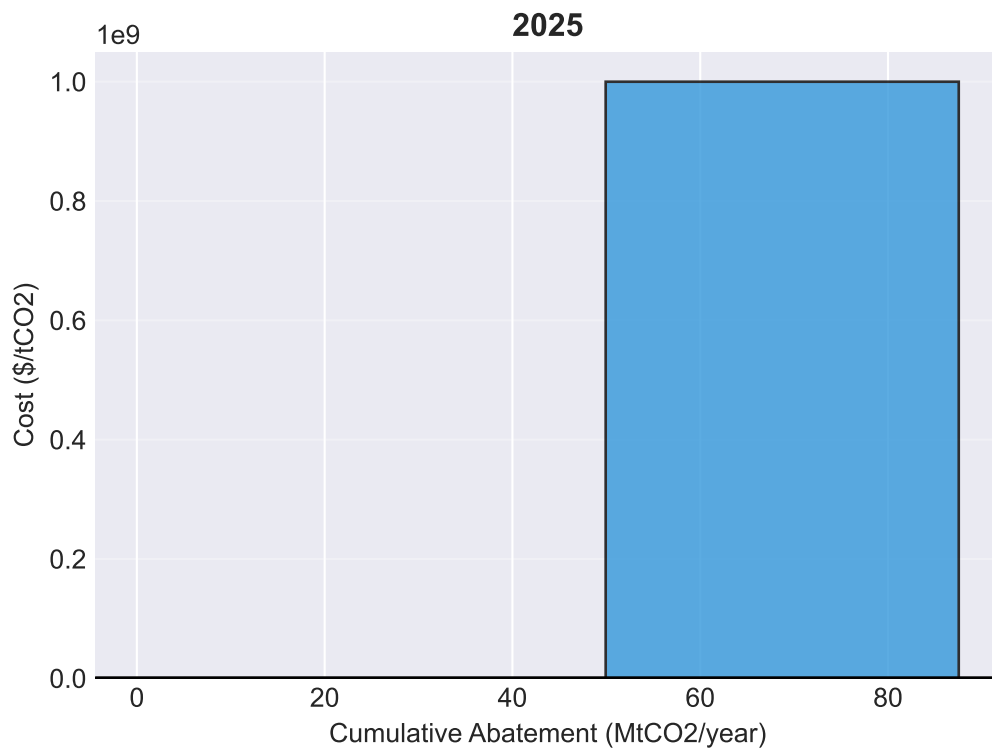
## Emission Reduction by Scenario



## Technology Deployment - Moderate Scenario

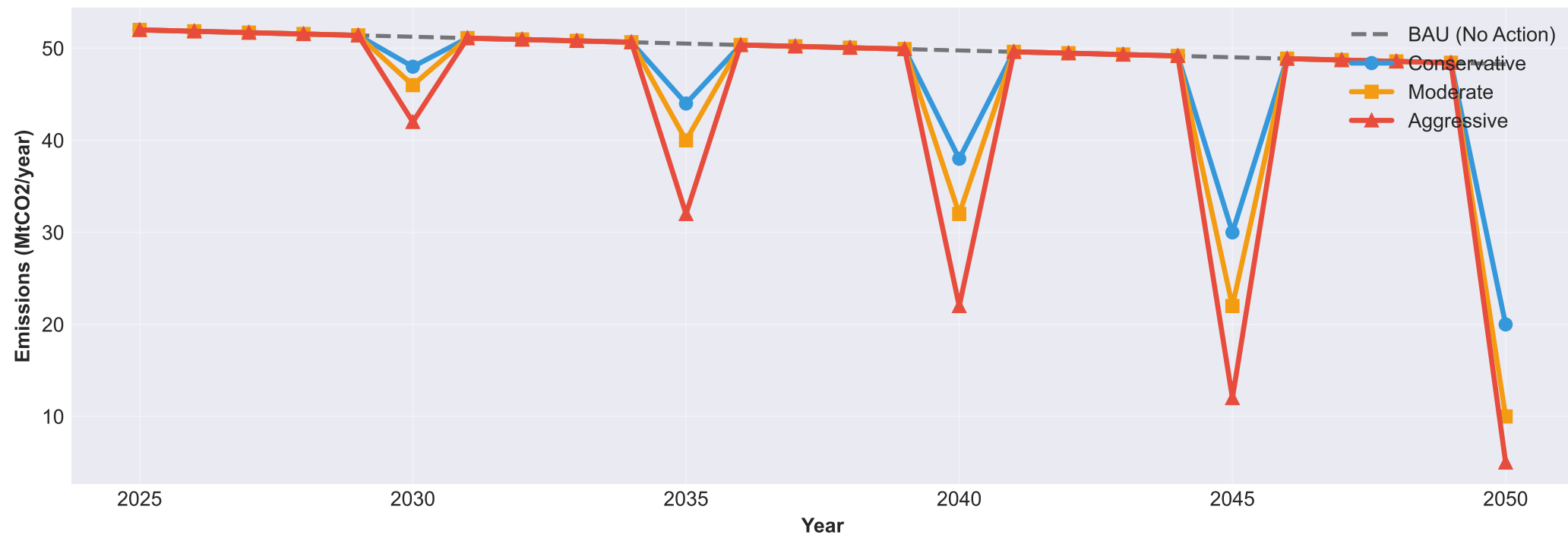


# MACC Curve Evolution (2025-2050)

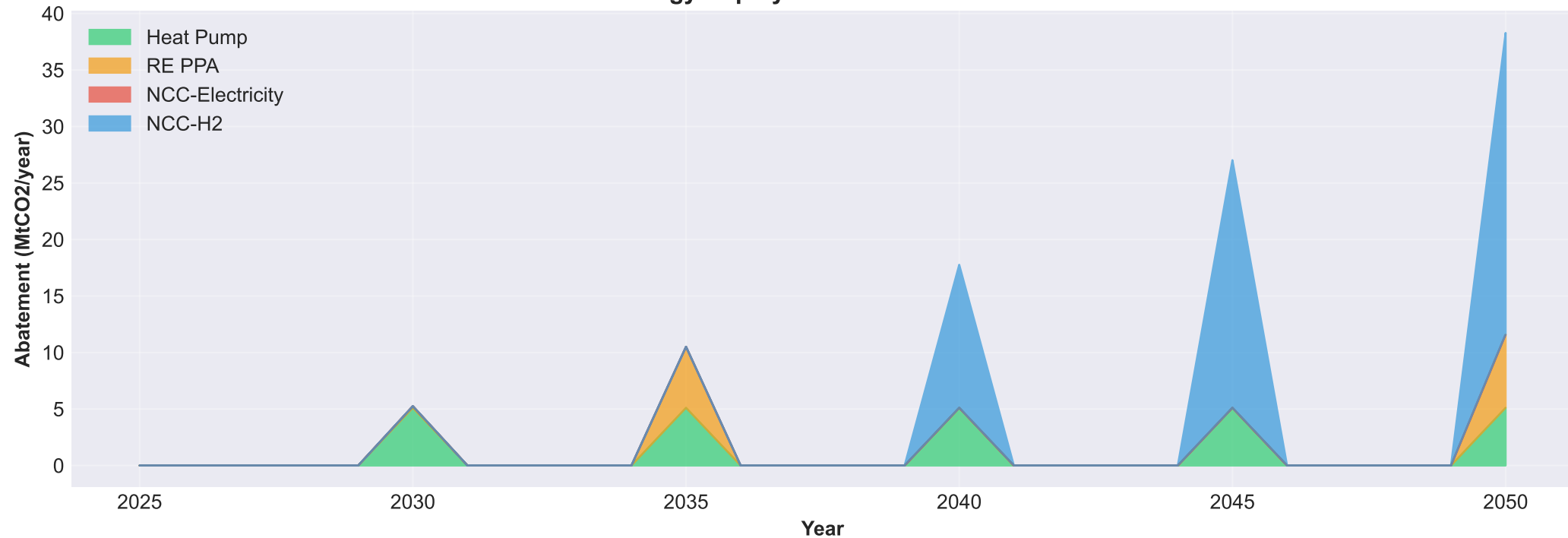


# Emission Trajectories by Scenario

## Annual Emissions



## Technology Deployment - Moderate Scenario



Summary Tables

Scenario Comparison

| Scenario     | 2030 Emis<br>(Mt) | 2030 Reduct<br>(%) | 2050 Emis<br>(Mt) | 2050 Reduct<br>(%) |
|--------------|-------------------|--------------------|-------------------|--------------------|
| Conservative | 48.0              | 7.7%               | 20.0              | 61.5%              |
| Moderate     | 46.0              | 11.5%              | 10.0              | 80.8%              |
| Aggressive   | 42.0              | 19.2%              | 5.0               | 90.4%              |

Technology Costs - 2030

| Technology      | Abatement<br>(MtCO2) | CAPEX+OPEX<br>(\$/tCO2) | Fuel Diff<br>(\$/tCO2) | Total Cost<br>(\$/tCO2) |
|-----------------|----------------------|-------------------------|------------------------|-------------------------|
| Heat_Pump       | 5.11                 | \$12.6                  | \$-760.6               | \$-748.0                |
| NCC-H2          | 37.60                | \$24.0                  | \$0.0                  | \$18.3                  |
| NCC-Electricity | 37.60                | \$28.7                  | \$0.0                  | \$-111.9                |
| RE_PPA          | 7.13                 | \$0.0                   | \$-131.1               | \$-131.1                |